

Impact of a Collaborative Care Model on Depression in a Primary Care Setting: A Randomized Controlled Trial

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To measure the effects of a collaborative care model that emphasized the role of clinical pharmacists in providing drug therapy management and treatment follow-up to patients with depression, we conducted a randomized controlled trial at a staff model health maintenance organization. We compared the outcomes of subjects treated in this collaborative care model (75 patients, intervention group) with subjects receiving usual care (50 patients, control group). After 6 months, the intervention group demonstrated a significantly higher drug adherence rate than that of the control group (67% vs 48%, odds ratio 2.17, 95% confidence interval 1.04–4.51, $p=0.038$). Patient satisfaction was significantly greater among members randomly assigned to pharmacists' services than among controls, and provider satisfaction surveys revealed high approval rates as well. Changes in resource utilization were favorable for the intervention group, but differences from the control group did not achieve statistical significance. Clinical improvement was noted in both groups, but the difference was not significant. Clinical pharmacists had a favorable effect on multiple aspects of patient care. Future studies of this model in other health care settings appear warranted.

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Revolutionary changes in the delivery and reimbursement of health care services have placed a greater emphasis on managing depressive illness in nonspecialty settings such as primary care.¹ Accumulated research evidence

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consistently has shown, however, that therapeutic end points for primary care patients with depression are rarely achieved.^{2,3}

Deficiencies have been noted in disease detection, treatment, follow-up, and referral to psychiatric services.^{4–7} These shortcomings can result in serious medical, functional, and economic consequences.^{8,9} Untreated depression, for instance, can predispose patients to developing other chronic medical conditions such as diabetes mellitus, cardiovascular disease, and Alzheimer's dementia.^{10–12} Also, patients with depression are heavy utilizers of health care resources, with direct medical costs that are approximately 3 times higher than those of patients without mental health disorders.¹³

For the management of depressive symptoms in primary care, antidepressant agents have emerged as a preferred treatment modality. Results of a large national survey revealed a 6-fold increase in the number of patients prescribed antidepressants between 1987 and 1997.¹⁴

Overall, 74% of all patients in the survey received antidepressants to treat their depressive symptoms. Although treatment guidelines advocate that patients receive at least 6 months of continuous pharmacotherapy, recent evidence suggests that only 30–40% will complete a full course.^{15–17} As inadequate antidepressant treatment has been associated with increased relapse rates and a comparatively poor prognosis, the consequences of premature discontinuation can be calamitous.¹⁸ Suboptimal treatment outcomes for primary care patients with depression have been attributed to severe limitations in provider time, inaccessibility to specialty services, and even to perceived deficiencies in medical training.^{1, 19, 20} Subsequent educational programs were implemented to improve the diagnosis and treatment of depression, but studies of this intervention have met with mixed results.^{21–23}

A more successful approach to improving outcomes in primary care integrates behavioral health experts.^{24, 25} Seminal studies show that psychiatrists and psychologists can have a favorable impact on patients with depression in this setting, particularly among patients satisfying the criteria for major depressive disorder who have comparatively severe symptoms.^{24, 26, 27} The cost-effectiveness of these treatment models among a more representative patient population (i.e., mild-to-moderate symptoms) has not been determined.^{28, 29}

The specialized skills of clinical pharmacists have proved to be beneficial for improving outcomes in a variety of health care settings.^{30–33} For instance, a recent position paper for the American College of Physicians–American Society of Internal Medicine expressed support for collaborative care arrangements involving pharmacists and physicians.³⁴ As the focus of depression treatment has turned to pharmacotherapy, institutions have created interdisciplinary treatment teams featuring clinical pharmacists in the role of consultant, educator, or care manager.³⁵ We previously described the results of a nonrandomized controlled study of an interdisciplinary primary care model in which clinical pharmacists worked under the supervision of psychiatrists to provide drug therapy management and patient education.³⁶ Results of this pilot project were favorable and included significant improvements in drug adherence, patient satisfaction, and resource utilization.³⁶ The purpose of the current investigation was to test the effects of this

particular collaborative care model on drug adherence rates, patient outcomes, provider and patient satisfaction, and medical resource utilization through a randomized controlled trial.

Methods

This investigation was conducted at Kaiser Permanente Medical Center in San Rafael, California, an upper middle class community approximately 20 miles north of San Francisco. The facility is part of a large nonprofit staff model and serves approximately 102,000 enrollees in the area. Thirty primary care providers were employed at the medical center and requested to refer prospective patients to the study coordinators if indicated.

The study protocol was approved by the Committee on Human Research at the University of California San Francisco Medical Center, as well as the investigational review board for Kaiser Permanente of Northern California. All subjects enrolled in the study were members of the health maintenance organization (HMO) who were receiving their primary care services at the San Rafael facility. Potential study subjects were referred to the protocol by their primary care provider immediately after starting antidepressant therapy for the expressed purpose of treating depressive symptoms. Physicians were instructed to exclude patients meeting the following criteria: evidence that subjects had received an antidepressant during the preceding 6 months; concurrent psychiatric or psychologic treatment; current symptoms of mania or bipolar disorder; psychotic symptoms; eminent suicidality; and active substance abuse or dependence. If psychiatric treatment was indicated at baseline or any time during the investigation, subjects were referred to the HMO's psychiatry department for care (or were permitted to self-refer) and subsequently were excluded from further study participation.

In most cases, patients were referred to the investigation and entered into the study on the day of their index visit. In an effort to replicate a naturalistic setting, the investigators did not use a structured diagnostic interview to determine study inclusion, but relied instead on the provider's clinical judgment that presenting symptoms warranted antidepressant treatment. The providers did not have any explicit incentives for study referrals. Subjects who returned the study surveys at end point were mailed a \$20 check as reimbursement for their

participation.

The co-principal investigators (P.R.F. and H.R.R.) served as depression care managers for the study subjects randomly assigned to the intervention group. Both investigators had previously obtained doctor of pharmacy degrees and had each accrued several years of direct patient care experience before study involvement. As one of the investigators was board certified in psychiatric pharmacy, he served as mentor for the other investigator during a 2-month training period that concluded just before the pilot project.

All study risks and procedures were explained to the study participants, and written informed consent was obtained before randomization. After the patients completed a brief survey to assess baseline depression severity (Brief Inventory for Depressive Symptoms [BIDS]) and functional impairment (Work and Social Disability Scale [WSDS]), the investigators opened a sealed envelope that determined study group assignment (intervention vs usual care). Subjects were randomly assigned to the collaborative care model or back to usual care in a 3:2 ratio. Subjects assigned to the usual care treatment arm received brief counseling on the prescribed drug, therapeutic end points, and side effects in a manner consistent with patient education routinely delivered to members receiving prescriptions from the HMO's outpatient pharmacy. The referring primary care provider was immediately notified if the subject was assigned to usual care, and subsequent treatment and follow-up were left to the provider's discretion. Patients in the intervention group were entered into the collaborative care treatment model described below.

Model of Care

The interdisciplinary treatment model under study in this investigation (known as the Medication Alliance Clinic) was implemented at the HMO facility approximately 2 years before the commencement of this randomized controlled trial. During the implementation phase, study investigators conducted a pilot project to assess the preliminary impact it would have on patient outcomes and satisfaction.³⁶ Initially, the collaborative care model was available only to a limited number of primary care providers (13 of 30 providers). Outcomes of patients referred by this cohort of physicians were compared with those of patients treated by

the facility's other primary care providers, and the results of this pilot project have been published.³⁶

The treatment protocol for patients randomly assigned to the intervention group during the randomized controlled trial was identical to that used during the pilot project. An intake interview was conducted by the care managers immediately after randomization. This clinic visit lasted approximately 30 minutes, and during this time, the care managers assessed the severity of psychopathology, identifying potential stressors and other predisposing factors. Medical, psychiatric, and drug therapy histories also were recorded, and care managers investigated whether any exclusion criteria were present.

Patient education was another important component of the intake interview and was emphasized at all other contact points as well. Symptoms, etiology, and prognosis of depression were discussed, and a detailed explanation of the role of antidepressants was presented (including potential therapeutic effects and adverse effects). Also, patients were advised of other treatment options and resources available at the medical center.

Under an institutional protocol approved by the medical center's Pharmacy and Therapeutics Committee, the care managers (or clinical pharmacists) were permitted to titrate antidepressant drugs in a manner consistent with the HMO's clinical practice guidelines and current recommended prescribing practices.¹⁵ The pharmacists were also allowed to prescribe ancillary drugs (e.g., trazodone for sleep), but if a change in antidepressant drugs was indicated, approval from the primary care provider was required.

After the initial intake interview, patients in the intervention group were scheduled for frequent follow-up contacts that consisted of both telephone calls and clinic appointments. The timing and nature of these contacts were designed to be consistent with key junctures in the recovery process, as well as with national treatment guidelines (Health Plan Employer Data Information Set [HEDIS]).³⁷ Telephone calls lasted 5–10 minutes and were scheduled for weeks 1, 2, 4, 10, and 16. During these calls, pharmacists followed a standardized set of questions that assessed drug adherence, therapeutic effects, adverse effects, and other social or medical factors. As clinical response became apparent, the pharmacists identified activities that the patients neglected during their illness and provided encouragement to resume

these pursuits (as evidence that they had achieved remission).

Brief clinic visits also were scheduled for weeks 6 and 24. As 6 weeks is considered to be the duration of a therapeutic trial, pharmacists evaluated clinical progress during this face-to-face encounter. At week 24, patients returned for their final visit, during which time the necessity of maintenance treatment was determined. Patients were advised to contact the clinic if they were considering the discontinuation of antidepressants at any time in the future.

Documentation of all patient contacts was entered into the official medical record in the form of a detailed progress note. If changes to the antidepressant regimen were warranted, the pharmacists communicated this recommendation to the provider by telephone call or through the facility's Lotus Notes messaging system. After the final clinic visit at week 24, a comprehensive summary of the treatment course and patient disposition was entered into the chart.

One of the more pivotal aspects of this treatment model was the supervision and involvement of a designated psychiatric mentor. Clinical pharmacists met with the psychiatrist for approximately 1 hour each week, summarizing the presentation of new patients, as well as providing updates on the clinical progress of other subjects. The psychiatrist was also available for consultation with the pharmacists during clinic hours (9 A.M. to 5 P.M., Monday–Friday) on an as-needed basis.

Outcome Measures

Adherence to antidepressant drug therapy was determined from the HMO's computerized prescription refill records. Measures of drug adherence were expressed as a medication possession ratio (MPR) or in terms of compliance with HEDIS specifications.³⁷ The MPR was defined as the number of days' supply of drug that the patient received during the 6-month study period, incorporating the quantity and strength of drug as well as prescribing directions. The MPR and HEDIS compliance values reflected drug adherence to all antidepressants received, including any drug changes or switches that may have occurred. The MPR values ranged from 0.167 (i.e., 1 month's supply during 6-month study period) to 1.0. For study purposes, full drug adherence was defined as an MPR value of 0.83 or more during the 6-month follow-up period (i.e., minimum of 5 months' supply of

antidepressant drugs dispensed). Within the context of HEDIS specifications, subjects were assessed for compliance within the early treatment phase (defined as at least 84 days' supply of drug during the first 114 days of treatment) and the continuation treatment phase (minimum of 180 treatment days during the 231-day study period).³⁷

Clinical and functional severity was assessed just before randomization (baseline) and again at 6 months through a mailed survey. Patients were asked to complete the BIDS, a validated 14-item patient-rated survey that ranks the severity of symptoms on a 4-point scale (range 0–42), at these two time points.³⁸ Both study groups also completed the WSDS, a 1-item, 5-point scale used to assess the degree of disability ranging from absent to severe.³⁹ Patient satisfaction was evaluated at month 6 (along with clinical and functional outcomes) through a 14-item survey mailed to both study groups.⁴⁰ The first 10 items asked patients to assess their satisfaction with various aspects of treatment on a 5-point scale, with responses ranging from very satisfied to very dissatisfied. The eleventh item asked patients if they would recommend Kaiser to a friend on a 4-point scale, with responses ranging from definitely yes to definitely no. Three other items on the satisfaction survey were presented in a dichotomous format (yes or no) inquiring into previous antidepressant treatment, previous psychotherapy, and whether or not the subject successfully completed the 6-month antidepressant treatment period.

Resource utilization was analyzed for the intervention and control groups through a comparison of the change in all clinic or emergency department visits between the 12 months preceding randomization and the 12 months immediately after. Patient visits to primary care providers, psychiatry, emergency department(s), and other specialty referrals were captured in the institution's comprehensive medical information system. As a preliminary survey of study patients revealed that referral outside of the closed staff model system was a rare phenomenon, these data were not included in the utilization analysis. Drug costs were captured in the automated prescription record and compared between study groups. Individual cost analyses were conducted and included the 6-month cost of the antidepressant regimen, as well as all other psychotropic drugs received (including antidepressants, antipsychotics, antianxiety compounds, and mood stabilizers)

Table 1. Patient Demographics at Baseline

Variable	Control Group (n=50)	Intervention Group (n=75)	p Value ^a
Female patients, no. (%)	42 (84)	64 (85)	0.84
Age, yrs (mean $\pm$ SD)	54.1 $\pm$ 17.3	54.4 $\pm$ 14.1	0.93
Chronic disease score (mean $\pm$ SD)	664.3 $\pm$ 428.4	606.5 $\pm$ 363.8	0.42
Patients starting SSRI therapy, no. (%)	48 (96)	66 (88)	0.12
Baseline BIDS score (mean $\pm$ SD)	18.3 $\pm$ 5.8	18.7 $\pm$ 5.8	0.73
Baseline WSDS score (mean $\pm$ SD)	1.9 $\pm$ 1.1	2.1 $\pm$ 0.9	0.26

SSRI = selective serotonin reuptake inhibitor; BIDS = Brief Inventory for Depressive Symptoms; WSDS = Work and Social Disability Scale.

^a χ^2 test.

and all drugs prescribed.

The experience and satisfaction of primary care providers with the collaborative care model was assessed in anonymous fashion through the one time distribution of a survey to all referring physicians. The nine-item survey was created in a format similar to that mailed to patients (i.e., five potential responses ranging from strongly agree to strongly disagree). Providers received a self-addressed envelope for survey return.

Statistical Analysis

Demographic variables and other baseline values were compared between the intervention group and study group. For these values and other outcome measures, a Student's *t* test was used for continuous data and a χ^2 analysis was conducted for categorical or dichotomous data. Statistically significant differences were defined a priori as a *p* value less than 0.05. In an effort to compare the total medical disease burden existing at baseline between groups, a chronic disease score was tabulated from the subjects' *International Classification of Diseases, Ninth Revision*, codes and prescription drug data.⁴¹

For statistical comparison of outcome measures, all data were analyzed initially for normal distribution, and appropriate statistical tests were used thereafter (as above). Potential differences were reported in terms of statistical significance (*p* value), as well as odds ratio (OR) when applicable. Drug adherence values (MPR and HEDIS compliance) were compared between study groups in an intent-to-treat fashion, inclusive of all patients receiving at least one prescription for antidepressants. Clinical and functional outcomes were compared as population means obtained at baseline and 6 months, as well as the change in population means occurring between the two time points. For all items on the patient satisfaction survey

consisting of a 5-point scale, data were compared by using a Student's *t* test, a χ^2 analysis of collapsed data (i.e., satisfied vs dissatisfied or neither), and a Wilcoxon score of ranked sums (nonparametric analysis). Changes in resource utilization were investigated by comparing the differences in population means evident for each type of service use. Resource utilization was also analyzed and compared as a dichotomous variable reflecting whether or not any service use had occurred in specific categories for individual subjects.

The sample size was set at a total of 200 randomized subjects, with an estimated effect size of 0.25 (i.e., 6-month drug adherence rates 25% higher in the intervention group). This calculation was based on a type I error rate of 5% (two-tailed), 90% statistical power, and an attrition rate of approximately 30%.

Results

A total of 125 patients were randomized into the study during the 15-month recruitment phase: 75 into the intervention group and 50 into the control group. Among patients referred to the investigators during this time, the most common reason for not participating was that an antidepressant had not yet been prescribed (e.g., primary care provider referred a patient to the clinic for further evaluation and/or treatment recommendation).

Most patients were female (85% in the intervention group, 84% among controls), and no statistically significant differences were noted between groups in age, medical comorbidity, functional impairments, depression severity, or class and dose of antidepressant (Table 1). Overall, study subjects appeared to have depressive symptoms best described as moderate (mean baseline BIDS score 18.5). Most had never received an antidepressant before (24%

with previous treatment in the intervention group vs 18% in the control group, $p=0.054$). Among the 75 patients assigned to collaborative care, 59 (79%) returned the mailed surveys, whereas 25 (50%) of the 50 control patients completed and returned these forms. Provider satisfaction surveys were returned by 18 of 27 referring physicians (at the end of the study, three providers were no longer working at this facility).

An intent-to-treat analysis of prescription refill records revealed that drug adherence rates were higher in the intervention group than among subjects randomly assigned to usual care (Figure 1). According to HEDIS guidelines for successful antidepressant treatment,³⁷ 57 patients (76%) in the intervention group were compliant with the early phase of treatment (vs 30 [60%] in the control group, OR 2.11, 95% confidence interval [CI] 0.97–4.58, $p=0.057$) and 50 patients (67%) completed the continuation phase (vs 24 [48%] in the control group, OR 2.17, CI 1.04–4.51, $p=0.038$). The MPR was higher for the intervention group than for the control group at both 3 months (0.92 vs 0.89, $p=0.48$) and 6 months (0.83 vs 0.77, $p=0.26$), but the difference did not achieve statistical significance. Patients in the intervention group were also more likely to change antidepressants during the study period (14 [19%] in the intervention group vs 2 [4%] in the control group, $p=0.016$). Rates of benzodiazepine therapy for the intervention and control groups were 20% (15 patients) and 28%

(14 patients), respectively ($p=0.299$). Maintenance doses of selective serotonin reuptake inhibitor antidepressants were comparable between groups.

Analysis of resource utilization patterns revealed that the patients randomly assigned to the collaborative care model experienced a 15% decrease in the total number of primary care visits, whereas the group receiving usual care had a 2% decrease ($p=0.14$ Student's t test between-group differences; Figure 2). The number of patients seeking emergency department visits increased slightly in the intervention group (7% increase in visits) and more dramatically in the usual care group (119% increase in visits, $p=0.10$ Student's t test). A nonsignificant increase in utilization of psychiatric services was recorded for both study groups during the 12 months after randomization ($p=0.66$). Overall, resource utilization increased slightly in the intervention group (5% increase in visits) and to a greater extent in the control group (24% increase in visits), a difference that was not statistically significant ($p=0.54$).

The institutional cost of drug regimens during the 6-month follow-up period was approximately 42% higher for the patients in the intervention group (vs that for the control group), but the wide range of individual patient values prevented this difference from being statistically significant ($p=0.18$). This same trend was noted in the cost of antidepressant regimens, as well as the cost of psychotropic drugs overall ($p=0.169$ and $p=0.206$, respectively). As differences in drug costs between groups may have been due largely to greater adherence rates among patients in the

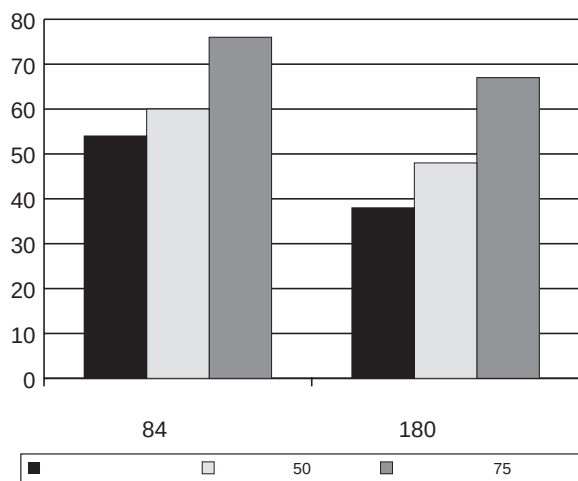


Figure 1. Drug adherence rates, as defined by Health Plan Employer Data Information Set specifications.³⁷ *The intervention group demonstrated a significantly higher drug adherence rate than that of the control group ($p<0.05$).

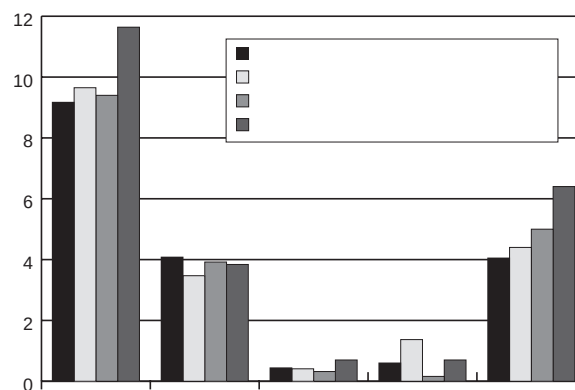


Figure 2. Changes in resource utilization. Mean number of visits/patient 12 months before and after randomization.

Table 2. Clinical Outcomes

Measure	Control Group (n=50)		Intervention Group (n=75)		p Value
	Value	No. of Surveys Evaluated	Value	No. of Surveys Evaluated	
Brief Inventory for Depressive Symptoms					
Baseline score, mean ± SD	18.3 ± 5.8	37 ^a	18.7 ± 5.8	75	0.73
Change in score at 6 mo, mean ± SD	-8.9 ± 8.3	24 ^b	-6.6 ± 7.3	54 ^b	0.23
Percentage with ≥ 50% reduction in score	54.1	24 ^b	40.7	54 ^b	0.27
Percentage achieving remission (score < 9)	58.3	24 ^b	55.6	54 ^b	0.36
Work and Social Disability Scale					
Baseline score, mean ± SD	1.9 ± 1.1	37 ^a	2.1 ± 0.9	75	0.26
Percentage with reduction in score	67	24 ^b	56	54 ^b	0.36

^aBaseline survey results were not available for the first 13 subjects.

^bOne of the 25 surveys returned in the control group and five of the 59 surveys returned in the intervention group were incomplete.

intervention group, a subsequent analysis of drug cost divided by MPR values was conducted and differences between study groups were found to be minimal ($p=0.511$).

Clinical and functional outcomes and patient satisfaction were determined from surveys mailed to all patients 6 months after randomization (Table 2). From survey responses, no statistically significant differences were found between study groups in clinical or functional outcomes. Overall, there was a 39% decrease in BIDS scores for the intervention group and a 49% decrease in scores for control subjects ($p=0.231$). Similar nonsignificant trends were noted in the percentage of patients achieving remission and those exhibiting a therapeutic response. Functional outcomes (evident from WSDS scores) indicated that 56% of the patients in the intervention group who returned the survey experienced an improvement in their condition and 67% of the control patients who returned a survey had the same benefit ($p=0.357$).

Results of the patient satisfaction survey revealed a much greater degree of treatment satisfaction for subjects randomly assigned to the collaborative care model than for control patients (Table 3). Responses to the survey were greater numerically (i.e., superior) for all 11 items addressing satisfaction, and statistically significant differences were found for six of these measures (nonparametric analysis). Specifically, patients in the intervention group expressed greater satisfaction than did control patients with the personal nature of care, availability of providers, ability of providers to listen, explanation of why antidepressants were prescribed, explanation of how to take the antidepressants, and patient's overall satisfaction with the HMO ($p<0.05$ for all measures, Wilcoxon scores of

ranked sums).

Survey results returned by providers were very positive, conveying the physician's satisfaction with workflow, patient welfare, and the pharmacists' abilities (Figure 3).

Discussion

The collaborative care model emphasizing the role of clinical pharmacists in managing therapy of patients with depression in a primary care setting was found to increase drug adherence rates, enhance patient satisfaction, and decrease the number of primary care visits. In addition, results of the provider satisfaction survey determined that primary care physicians were very pleased with the intervention and thought that the collaborative care model enabled them to increase productivity. Collectively, these results suggest that the implementation of this service improved the quality of patient care and had a favorable impact on institutional resources.

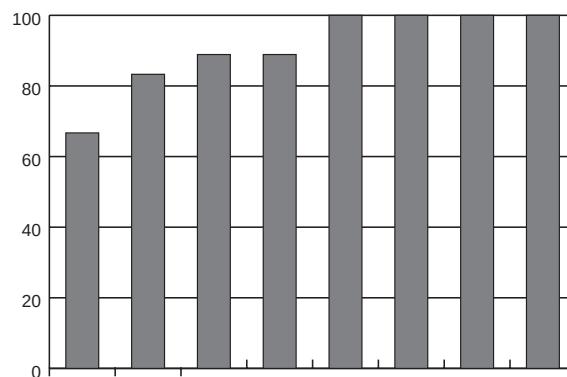


Figure 3. Provider satisfaction survey (18 respondents).

Table 3. Patient Satisfaction Survey Results

Item	5-Point Scale ^a Results				p Value ^b	p Value ^c	p Value ^d
	Control Group (n=33)		Intervention Group (n=59)				
	Mean	SD	Mean	SD			
1.The overall treatment of your depression.	3.82	0.81	4.15	0.81	0.060	0.066	0.023
2.The personal nature of the care you received for this condition.	4.00	0.86	4.39	0.75	0.029	0.035	0.032
3.The ability of the Kaiser personnel to listen and acknowledge your concerns.	4.18	0.81	4.53	0.63	0.026	0.016	0.037
4.The explanation about why the antidepressants were prescribed.	3.94	0.79	4.33	0.71	0.018	0.154	0.018
5.The information you were given about how to take your antidepressant medication.	4.33	0.60	4.58	0.56	0.055	0.547	0.049
6.The information you were given on what to expect and how to deal with side effects.	3.94	0.89	4.25	0.79	0.095	0.089	0.101
7.The effectiveness of the medication prescribed.	3.56	1.37	3.68	1.18	0.676	0.697	0.848
8.Availability of advice for this condition.	3.79	0.93	4.26	0.88	0.017	0.026	0.001
9.Kaiser's follow-up with you for this condition.	3.53	1.11	4.29	0.84	<0.001	0.004	0.116
10.Kaiser Permanente, overall.	3.94	0.83	4.18	0.83	0.193	0.090	0.044
11.If a friend were in need of similar help, would you recommend Kaiser Permanente? ^e	3.15	0.71	3.39	0.70	0.132	0.189	0.104
Dichotomous Format Results:							
	Yes Response (%)						
Item	Control Group (n=33)		Intervention Group (n=59)			p Value ^c	p Value ^d
12.Did you complete 6 months of treatment?	65.63		79.66			0.141	0.145
13.Have you ever received antidepressant medications before?	18.18		23.73			0.536	0.542
14.Have you received any psychological counseling during the last 12 months?	33.33		30.51			0.780	0.784

^a1 = very dissatisfied; 2 = dissatisfied; 3 = neither satisfied nor dissatisfied; 4 = satisfied; 5 = very satisfied.

^bStudent's *t* test.

^c χ^2 test paired data.

^dWilcoxon score (rank sums).

^eResponse format was a 4-point scale: 1 = yes, definitely; 2 = yes, I think so; 3 = no, I don't think so; 4 = no, definitely not.

Attempts to demonstrate an improvement in clinical and functional outcomes with this intervention were unsuccessful. Investigators found that both groups responded well to treatment and no statistically significant differences were forthcoming. This finding can be attributed, in part, to the limited number of subjects randomized into the trial. Recruitment rates for study randomization diminished over time as providers insisted that patients be enrolled in the collaborative care model (i.e., providers chose not to "risk" randomization). Also, survey return rates were substantially lower in the control group than in the intervention group, further limiting the power of this study to detect a statistically significant difference in clinical outcomes. The limited number of returned surveys also raises the possibility that patients with better clinical outcomes in this cohort were more likely to respond (i.e., selection bias).

A comparison of the results from this randomized controlled trial with those of a similar controlled investigation (pilot project) revealed several consistent findings.³⁶ Both studies of this model, for instance, reported a significant increase in drug adherence rates and a greater reduction in primary care visits during the 12-month study period for the intervention group versus the control group. Responses to patient satisfaction surveys were also very favorable in both intervention groups, and many of the same items achieved a statistically significant difference from controls (e.g., treatment follow-up, provider accessibility, overall satisfaction with HMO).

The MPR among patients in the collaborative care model was quite high in the randomized controlled trial and pilot project (0.84 and 0.81, respectively), indicating that a strong majority of patients completed 6 months of antidepressant treatment. Pooled data from the two investigations

(166 patients) in intervention groups reveal that 78% of all patients referred to the clinical pharmacists in the collaborative care model achieved national treatment guideline standards. This is nearly a 2-fold increase over recent national benchmark figures for the continuation phase.¹⁷ As the completion of a full course of pharmacotherapy has been shown to ameliorate depressive symptoms in the short term, as well as lower the likelihood of future episodes, the benefits of drug adherence may persist long after the 6-month treatment phase.¹⁸ For instance, longitudinal investigations have demonstrated that the risk of relapse increases in a nonlinear fashion with each successive episode.⁴²⁻⁴⁴ The successful completion of therapy, therefore, may impart benefits to the patient and managed care organization for years after the index episode.

Of interest, the drug adherence rate of control subjects was substantially higher in the randomized controlled trial (MPR = 0.77) than it was during the pilot project immediately preceding (MPR = 0.66).³⁶ One potential explanation for this finding is that physician prescribing practices and follow-up improved significantly after the establishment of the pharmacists' clinical services at the medical center (i.e., interventional bias). Similarly, the contact that investigators had with patients at baseline (e.g., describing investigation, distributing depression surveys, providing routine patient education) may have had a favorable impact on the patient's familiarity with the disease and comfort with accessing the health care system. Alternatively, the quality of care received by patients in the control group may have been enhanced by the providers' awareness of the ongoing investigation (i.e., Hawthorne effect). As compliance rates for the control group during the randomized controlled trial were considerably higher than national benchmark figures (48% vs 38%, HEDIS specifications for continuation phase treatment), one may conjecture that patients in this investigation did receive a relatively superior level of care.¹⁷

The reported success of this collaborative care model may be attributed to several factors. For instance, as greater emphasis is placed on prescription drugs to treat psychiatric disorders (vs psychotherapy or other interventions), the educational background and training of clinical pharmacists may be of particular value in improving treatment outcomes among the mentally ill. This benefit was recently demonstrated in a prospective study conducted

in the community pharmacy setting where pharmacist monitoring was correlated with greater patient satisfaction and antidepressant adherence.⁴⁵ Prescribing patterns evident from the present randomized controlled trial (and previous pilot project) demonstrated that clinical pharmacists tended to treat patients with comparatively low doses of antidepressants and to slowly advance these doses if necessary, improving patient acceptance of psychotropic treatments in primary care. Switch rates of antidepressant drugs were significantly higher for patients treated in the collaborative care model (vs controls), which may serve as a surrogate measure for the intensity of treatment follow-up, as well as an indication that drug selection was more closely tailored to fit individual patient needs.

Another potential benefit of this treatment model may be seen in the timing and nature of patient contacts (clinic visits vs telephone calls). Clinic visits at baseline, 6 weeks, and 24 weeks enabled the pharmacists to forge a strong therapeutic alliance with patients that may not have been achieved if telephone contacts were the exclusive form of follow-up.^{40, 46} Face-to-face contacts with patients from day 1 enabled pharmacists to achieve a greater degree of familiarity with patients and also allowed them to witness changes in illness severity that may not be apparent over the telephone.

The role of psychiatric liaisons or mentors in the success of this program should also be acknowledged. Weekly case conferences with pharmacists and a designated psychiatrist served to clarify diagnostic issues, as well as more clearly delineate treatment plans. Psychiatrists were also available for off-site telephone consultation on an as-needed basis for more pertinent issues. Numerically, a greater increase in psychiatric referral rates was found among patients in the control group ($p=0.66$). However, the general perception among psychiatrists at the medical center was that the intervention model managed to better streamline specialty services and discourage unnecessary referrals.

Accurate interpretation of the results of this investigation requires acknowledgment of the following study limitations: the study was not double-blinded (i.e., subjects and providers were presumably aware of study group assignments after randomization); the patient population consisted primarily of older female patients residing in an affluent suburban community, and results may not necessarily be generalizable to

other demographic groups; educational background information was not available in the HMO database; the collaborative care model was staffed by a limited number of clinical pharmacists (i.e., two pharmacists), and results may not necessarily be reproducible in other settings; this treatment model was created and implemented within the confines of a closed staff model, and prospects for implementing this service in other health care settings (i.e., integrated health plan networks) are unknown; and study methodology was not adequate to determine which aspect of the intervention was uniquely responsible for improved outcomes.

The size and scope of this investigation did not permit the investigators to thoroughly examine the cost-effectiveness of this service. A comparison of resource utilization patterns for the 12 months preceding randomization and the 12 months immediately after uncovered a greater decline in the number of primary care and emergency department visits among patients in the intervention group than in the control group ($p=0.10$ and 0.14 , respectively). While the total number of visits to primary care, emergency department, psychiatry, or other medical specialties increased to a greater extent in the control group, this difference was not statistically significant and the practical importance of this cost savings is unknown ($p=0.54$).

The institutional cost of providing these services can be attributed primarily to the reimbursement of clinical pharmacists' time. In this particular model, pharmacists spent an average of 71 minutes in direct contact with each patient over the course of the 6-month protocol (based on 6.0 successful telephone calls averaging 7 min and 1.6 office visits averaging 20 min). Other costs that may not have been quantified but should be considered in the economic analyses include those of psychiatrists' time, institutional overhead, pharmacists' benefits, time required for pharmacists' training, unsuccessful telephone calls, and the documentation of services. As the primary cost factor driving similar collaborative care models was that of the care manager, it seems unlikely that the comprehensive cost of this particular model would approach others previously reported, but further study is warranted to confirm this.^{47, 48} Similarly, additional study of this collaborative care model may help to define specific subpopulations of patients with depression who may benefit from this approach (i.e., those with other medical comorbidities).

From a health policy perspective, the results of this trial may have future implications. For instance, demonstrated improvements in drug adherence and other patient outcomes may suggest that skilled drug therapy management and frequent longitudinal follow-up by specialized personnel is vital to any alternative treatment models for depression. Similarly, the respective abilities of clinical pharmacists and psychiatrists appear to be well suited to future collaborative efforts, and the integration of their services into primary care settings may be economically feasible, as well as clinically effective.

Conclusions

The creation of a unique interdisciplinary treatment model emphasizing clinical pharmacy services was associated with significant increases in drug adherence, greater patient satisfaction, and improved resource utilization patterns among patients with depression who are treated in the primary care setting. A combination of skilled management of drug therapy and close treatment follow-up may have been chiefly responsible for these study findings. Future multisite investigations of this particular collaborative care model in other health care environments would appear to be prudent.

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